Duty at home

DM6 · Study of a logarithmic function · 55 min.

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{1+\ln x}{x}$. Its representative curve is noted $\mathcal{C}_f$.

1. Determining the limits of f at the boundaries of the domain, study his sign, then draw his picture changes.
2. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$.
3. Demonstrate that the equation $f(x)=\frac{1}{2}$ admits exactly two solutions. Give each one one justified framework using calculated values.
4. Using the maximum of f, demonstrate that $1+\ln x \leq x$ for every $x>0$. Specify the case equality.